

● EASY

● GEOMETRY AND TRIGONOMETRY

Geometry and Trigonometry

03

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1

GEOMETRY AND TRIGONOMETRY

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D , and angles C and F are right angles. If $\cos B = \frac{1}{22}$, what is the value of $\cos E$?

A. $\frac{1}{22}$

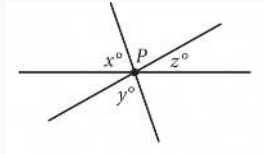
B. $\frac{1}{23}$

C. $\frac{21}{22}$

D. $\frac{22}{23}$

The side length of a square is 55 centimeters cm. What is the area, in cm^2 , of the square?

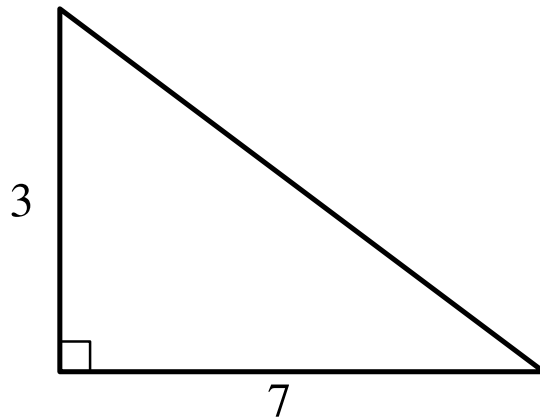
- A. 110
- B. 220
- C. 3,025
- D. 12,100



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

- A. 140
- B. 80
- C. 40
- D. 20



Note: Figure not drawn to scale.

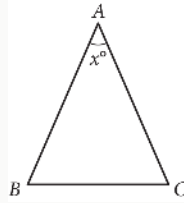
- One angle is a right angle.
- The lengths of the sides adjacent to the right angle are as follows:
 - 3
 - 7
- A note indicates the figure is not drawn to scale.

The lengths of the legs of a right triangle are shown. Which of the following is closest to the length of the triangle's hypotenuse?

- A. 3.2
- B. 5
- C. 7.6
- D. 20

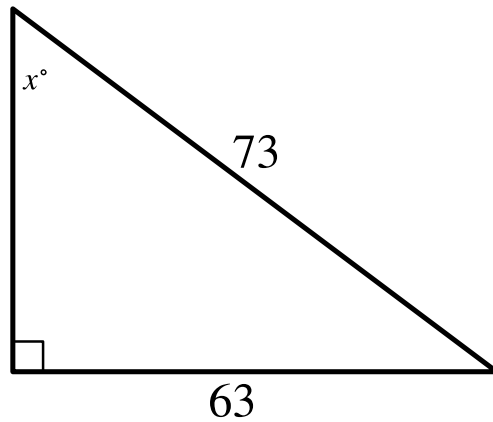
What is the area, in square centimeters, of a rectangle with a length of 36 centimeters and a width of 34 centimeters?

- A. 70
- B. 140
- C. 1,156
- D. 1,224



In the given triangle, $AB = AC$ and $\angle ABC$ has a measure of 67° . What is the value of x ?

- A. 36
- B. 46
- C. 58
- D. 70



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the side opposite the right angle is 73.
- The length of the side opposite the angle with measure x° is 63.
- A note indicates the figure is not drawn to scale.

In the right triangle shown, what is the value of $\sin x^\circ$?

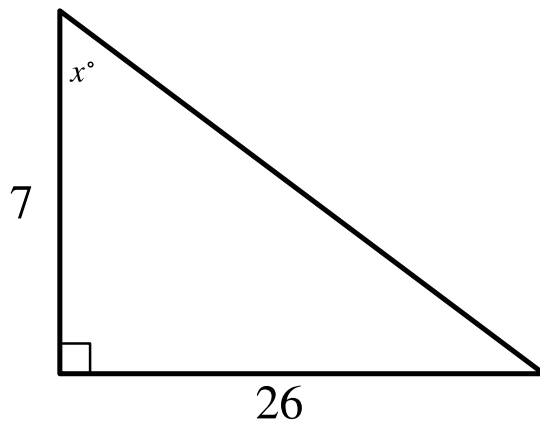
- A. $\frac{1}{73}$
- B. $\frac{10}{73}$
- C. $\frac{63}{73}$
- D. $\frac{136}{73}$

The perimeter of triangle ABC is 17 inches, the length of side AB is 4 inches, and the length of side AC is 7 inches. What is the length, in inches, of side BC ?

- A. 4
- B. 6
- C. 7
- D. 11

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

- A. 10
- B. 20
- C. 30
- D. 35



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is x° .
- The length of the leg opposite the angle with measure x° is 26.
- The length of the leg adjacent to the angle with measure x° is 7.
- A note indicates the figure is not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$